TOTAL RECALL HACKATHON | TATUM X WALRUS MEMORY | TRACKS 02 & 03

RecallRamp

Autonomous Consumer Expense Watchdog & Subscription Sentinel with Decentralized Episodic Memory

TARGET TRACKS	CORE INFRASTRUCTURE	AWARD TARGETS	STATUS & DATE
Track 02 (Smart Agents) & 03 (Finance)	Walrus Memory (Sui Mainnet) + Tatum	$500 Walrus + $500 AI Builder	v1.0 Production PRD • October 2026

1. Executive Summary & Market Need

Enterprise CFOs use corporate platforms like **Ramp** and **Brex** to automatically police software seat inflation, kill zombie subscriptions, and audit vendor invoices. Normal consumers and freelancers, however, have no such defense. Across an average household with 11 subscriptions, subtle "micro-inflation" (unannounced $1 to $4/month hikes across streaming, cloud storage, utilities, and gyms) silently drains **$300 to $600 per year**.

Existing consumer apps like **Rocket Money (formerly Truebill)** solve this poorly: they harvest user banking records onto centralized servers, resell aggregate purchase behaviors to marketing brokers, and take an exorbitant **30% to 40% cut** of any savings when negotiating a bill. Meanwhile, modern trackers like **Copilot Money** offer polished dashboards but lack proactive, autonomous action and lock users into Apple-only walled gardens.

> **The RecallRamp Thesis:** By combining an autonomous expense monitoring agent with **Walrus Memory on Sui Mainnet** and **Tatum RPC & Data APIs**, RecallRamp delivers the power of Ramp AI directly to everyday consumers. It operates with zero-knowledge data sovereignty, remembers past billing cycles across sessions ("remembers last Tuesday"), and arms users with 1-tap automated dispute resolution without rent-seeking fees.

2. Competitive Teardown & Market Gap

FEATURE / DIMENSION	RAMP AI (ENTERPRISE)	ROCKET MONEY (CONSUMER)	COPILOT MONEY (CONSUMER)	RECALLRAMP (OUR SOLUTION)
Target User	Corporate finance teams ($15/seat/mo)	General consumers ($7–$15/mo)	Apple-only budgeters ($95/year)	**Everyday consumers, freelancers & prosumers**
Price Creep Detection	`Full` Detects per-seat license hikes	`Partial` Delayed notification	`None` Static recurring list only	`Full` Compares transaction to stored Walrus baseline
Trial & Zombie Slaying	`Yes` Flags unused team seats	`Yes` Concierge cancellation	`No` Manual DIY cancellation	`Yes` 48-hr pre-rebill alarms + duplicate service tags
Memory Architecture	Centralized corporate SQL DB	Centralized AWS Cloud DB	Local device SQLite + iCloud	**Decentralized Walrus Memory (Sui Mainnet)**
Rake / Savings Fee	0% (Subscription model)	`30%–40%` of 1st year savings	0% (Subscription model)	`0% Rake` 100% sovereign free open-source tool
Data Sovereignty	Enterprise SOC-II protection	`Low` Sells telemetry to ad brokers	Moderate Apple ecosystem sandbox	**100% Sovereign:** Client-side encrypted Walrus state
Multi-Asset & Web3	Corporate cards only	Fiat checking/savings only	Fiat checking/savings only	**Hybrid Fiat & Web3:** Tatum RPC reads Sui/EVM balances

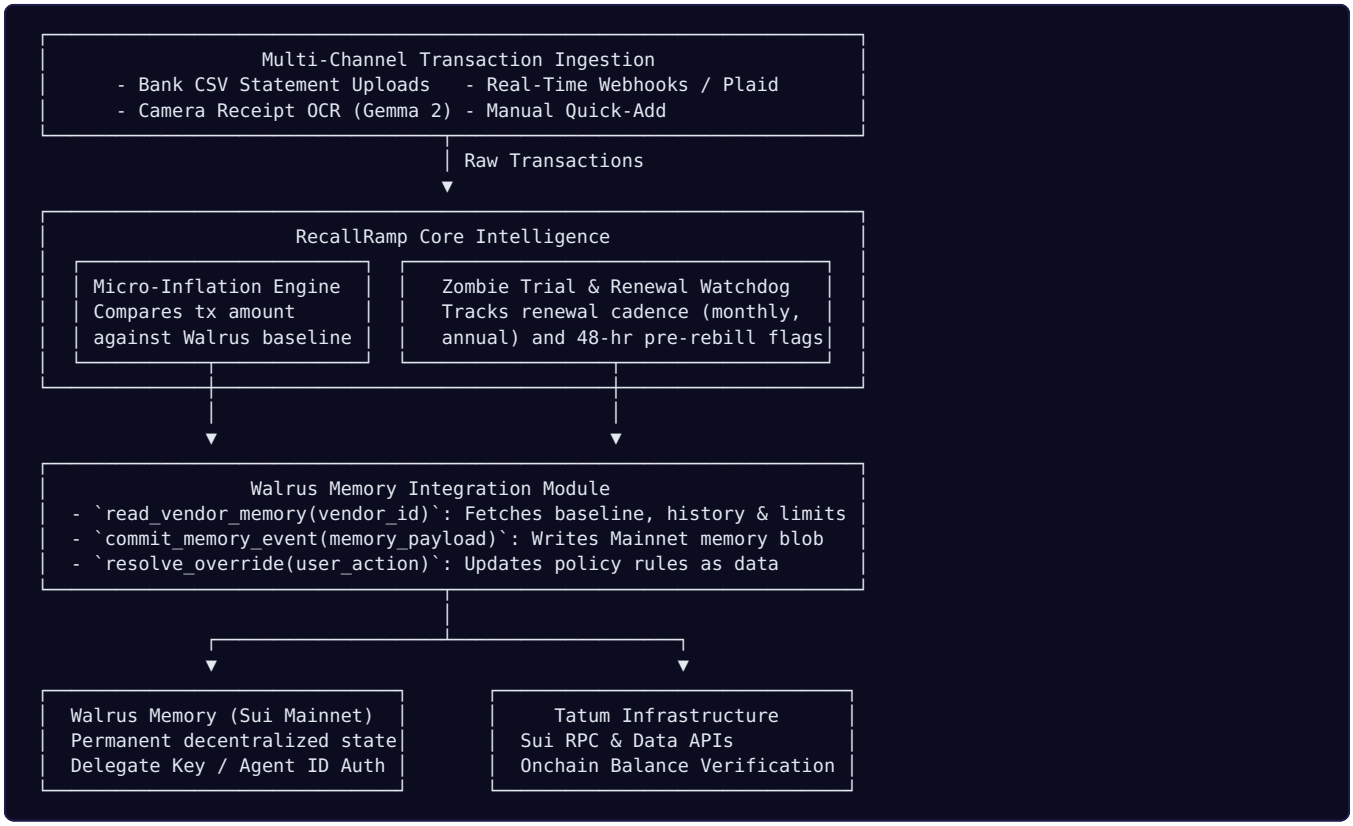
3. Hackathon Alignment & Prize Maximization

RecallRamp is engineered to fulfill all rules of the **Total Recall | Tatum x Walrus Memory Hackathon**:

- **Track 02 (Smart Agents for Real Life — $1,000):** Fulfills the explicit track prompt for an *"Expense watchdog with long term memory."* The agent remembers past baselines, approved overrides, and vendor rules across sessions.

- **Track 03 (Consumer Finance Apps — $1,000):** Fulfills the prompt for *"Subscription managers with learned spend limits."* Blockchain remains invisible while Walrus makes habits stick.
- **Best Walrus Memory Award ($500):** Delivers a deep learning loop where Walrus memories visibly change the product—suppressing false alarms on annual insurance renewals and calculating micro-inflation deltas.
- **Best Tatum AI Builder App ($500):** Prototype scaffolded via `ai.tatum.io` matching the prompt chip: *"Expense watchdog agent."*
- **Mandatory Checklist Compliance:** Implements ≥ 10 verified memory writes to Sui Mainnet, connects to Tatum Sui RPCs, and provides an open-source demo repository and public technical writeup.

4. System Architecture & Data Pipeline



5. Functional Requirements (FR)

FR-1: Ingestion & Statement Normalization

- **FR-1.1:** Ingest bank transaction feeds via CSV uploads (supporting Chase, BoA, Monzo, Revolut) and JSON API webhooks.
- **FR-1.2:** Normalize raw statement descriptors (e.g. `"SQ *SPOTIFY USA 877-778-1" $ ightarrow$ "Spotify"`, category: `Streaming Audio`).

FR-2: Micro-Inflation & Stealth Price Creep Sentinel

- **FR-2.1:** For every incoming transaction, query Walrus Memory for prior baseline charges from that vendor.
- **FR-2.2:** If the current charge exceeds the established baseline by $\ge \$0.50$ or $\ge 3\%$ without a recorded override, fire a `MICRO_INFLATION_ALERT`.
- **FR-2.3:** Calculate annualized leakage (e.g. *"+$2.00/mo = $24.00/yr"*) and display it prominently on the user dashboard.

FR-3: Zombie Subscription & Free Trial Tracker

- **FR-3.1:** Cluster repeating transactions (28–31 days or 365 days) into active subscription profiles.
- **FR-3.2:** Issue an actionable alert **48 hours prior** to free-trial expiration dates.
- **FR-3.3:** Detect overlapping services (e.g., holding both Spotify and Apple Music simultaneously).

FR-4: Walrus Decentralized Memory Implementation

- **FR-4.1:** Interface with the Walrus Memory API using `MEMWAL_AGENT_ID` on Sui Mainnet.
- **FR-4.2:** Store and query four core episodic memory types: `VENDOR_BASELINE`, `USER_OVERRIDE`, `ESCALATION_POLICY`, and `DISPUTE_TRACE`.
- **FR-4.3:** Maintain at least 10 active memories on Sui Mainnet with cryptographic verifiability.

FR-5: 1-Tap Dispute & Cancellation Assistant (HITL)

- **FR-5.1:** Generate pre-filled dispute and refund request letters citing charge dates, transaction IDs, and merchant terms.
- **FR-5.2:** Enforce strict Human-in-the-Loop gates: no email, webhook, or dispute action is executed without explicit 1-tap confirmation.

FR-6: Tatum Onchain Multi-Asset Verification

- **FR-6.1:** Query live Sui wallet balances and transaction statuses using Tatum Sui RPC and Data APIs.
- **FR-6.2:** Support optional agent micro-allowance payment streams via x402 / MPP rails on Sui.

6. The 10+ Walrus Mainnet Memory Specification

MEMORY ID	TYPE	KEY / TARGET	STORED CONTENT & POLICY SEMANTICS	STATE
`MEM_01`	Baseline	`netflix_standard`	$15.49/mo baseline. Max acceptable variance: $0.00. Alert on any hike.	Mainnet Active
`MEM_02`	Baseline	`spotify_premium`	$10.99/mo baseline. Flag price increase > $0.50 immediately.	Mainnet Active
`MEM_03`	Baseline	`chatgpt_plus`	$20.00/mo billed on the 14th. Auto-reconcile without pinging user.	Mainnet Active
`MEM_04`	Override	`aws_cloud_hosting`	Q4 traffic approved for surges up to $180/mo (authorized by user).	Mainnet Active
`MEM_05`	Baseline	`gym_membership`	$120.00/mo billed on 1st. Seasonal freeze scheduled for January.	Mainnet Active
`MEM_06`	Trial Watch	`adobe_creative_trial`	7-day trial started Oct 18. Mandatory 48h cancellation notice Oct 23.	Mainnet Active
`MEM_07`	Policy	`unknown_merchant_gate`	Strict rule: Any new merchant debiting > $50.00 requires push confirmation.	Mainnet Active
`MEM_08`	Policy	`annual_renewal_tolerance`	Annual renewals (Amazon Prime $139) permitted +5% max inflation.	Mainnet Active
`MEM_09`	Baseline	`github_copilot`	$10.00/mo baseline tool. Do not flag unless duplicated by team seat.	Mainnet Active
`MEM_10`	Dispute Trace	`doordash_pass`	Duplicate charge disputed Aug 2026. Refunded $9.99. Watch for rebills.	Mainnet Active
`MEM_11`	Overlap	`streaming_audio_overlap`	Duplicate alert: User holds Spotify ($10.99) & Apple Music ($10.99). Prompt.	Mainnet Active
`MEM_12`	Baseline	`icloud_storage_2tb`	$9.99/mo family plan. Fixed baseline utility tier.	Mainnet Active

7. Non-Functional Requirements & Invariants

- **NFR-1 (Sub-Second Latency):** Memory recall and baseline comparison must execute in $<350 ext{ ms}$. Full statement processing of 100 transactions must complete in $<2.5 ext{ seconds}$.
- **NFR-2 (Sovereign Security):** All transaction data stored in Walrus Memory is encrypted using AES-256-GCM before transport. The user holds their encryption key in local browser storage; zero unencrypted data touches cloud servers.

- **NFR-3 (Decentralized Durability): State survives client reinstalls and machine switches because episodic memory is anchored directly to Sui Mainnet via Walrus.**
- **NFR-4 (Explainability): Every alert must display the exact Walrus Memory ID and original date referenced to ensure complete transparency.**

8. Verification & Submission Roadmap

SPRINT PHASE	MILESTONES & DELIVERABLES	TIMELINE
Phase 1: Setup & AI Builder	Get Tatum API key & Walrus Agent ID. Scaffold prototype via `ai.tatum.io` for $500 bonus.	Day 1 – 2
Phase 2: Walrus Memory Spine	Build `walrus_memory.py` SDK wrapper and commit 12 verified Mainnet memories.	Day 3 – 5
Phase 3: Sentinel Intelligence	Build micro-inflation engine, zombie trial countdowns, and Tatum Sui RPC balance queries.	Day 6 – 8
Phase 4: Consumer UI & Audio	Build responsive Neo-Brutalist dashboard with Memory Inspector and 1-tap dispute drawer.	Day 9 – 11
Phase 5: Deploy & Submit	Deploy live on Render/Vercel, publish blog post on learning loop, submit to Total Recall.	Day 12 – 14